MBA, Analytics Capstone: Electric Vehicle Charging Infrastructure Expansion Proposal

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The Business

ChargePoint is a leading electric vehicle (EV) charging network provider in the United States, offering a comprehensive ecosystem of charging hardware, software, and services. The company operates one of the largest networks of independently owned EV charging stations, connecting drivers to a wide range of public and private charging locations.

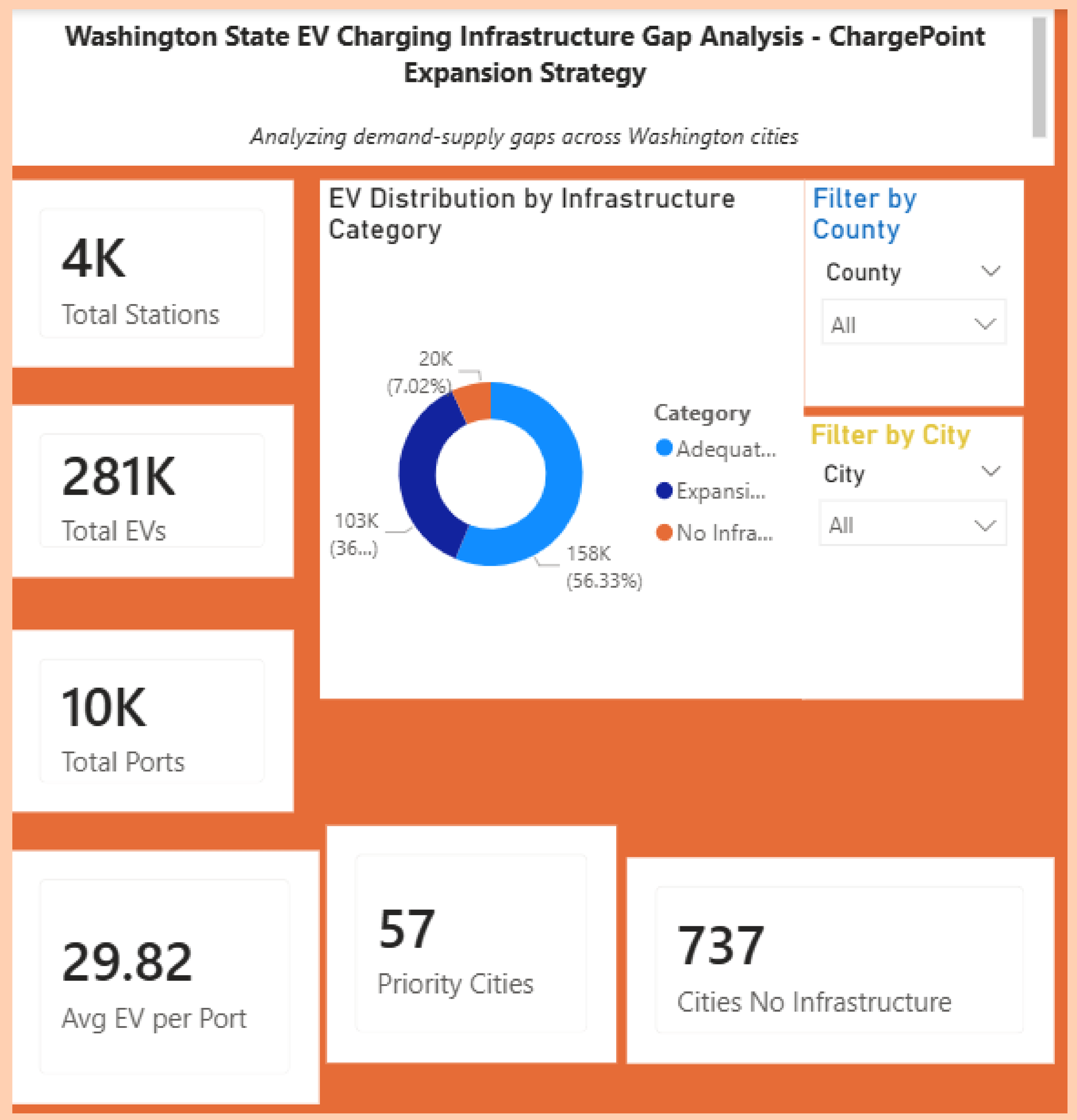
Objective & Goals

Objective: Identify optimal locations for EV charging infrastructure expansion using data-driven analysis.

Goals:

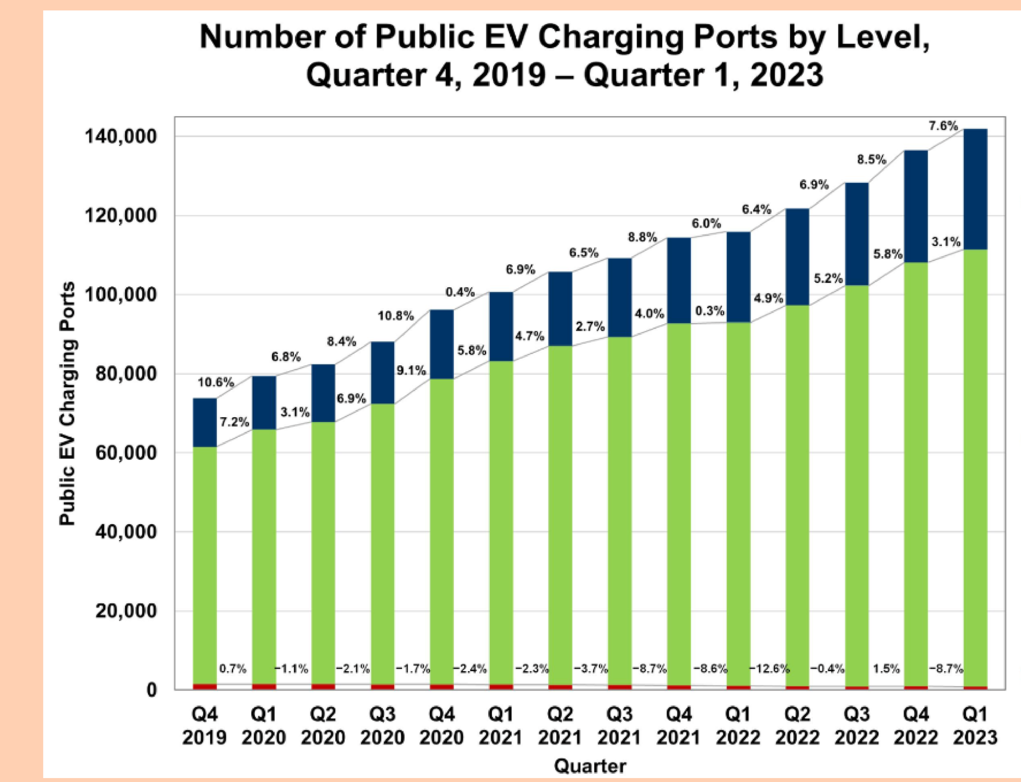
- Identify regions with high EV demand and limited charging infrastructure
- Analyze utilization patterns of existing charging stations
- Estimate potential return on investment (ROI) for infrastructure expansion
- Provide strategic recommendations for ChargePoint's growth and market positioning

Data Exploration



Dataset Overview:
EV Registrations (~280K) and Charging Infrastructure (3,009 stations, 8,404 ports) across WA State.
Key Variables:
EV_Registrations (demand), Total_Ports (supply), EVs_per_Port (gap/strain metric)
Data Cleaning:
Filtered to WA, nulls removed, standardized city names, aggregated and merged at City level.
Summary Statistics:
554 cities analyzed; ~56% have EVs but no charging infrastructure.
Trends & Visualizations:
EV demand is concentrated, infrastructure uneven: high EVs per port highlights underserved regions and expansion opportunities.

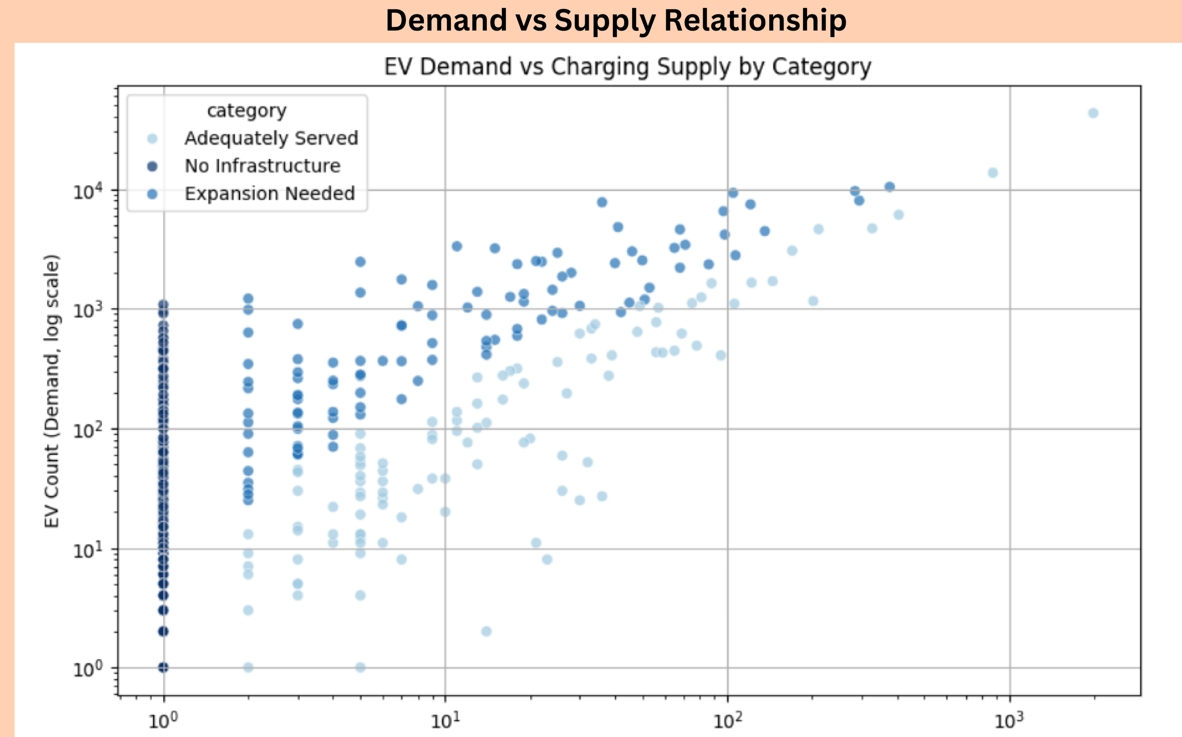
Market Analysis



Currently, infrastructure expansion is relatively stable and linear, while EV adoption is accelerating more quickly. This means there may not be enough stations to support the increasing number of EV users.

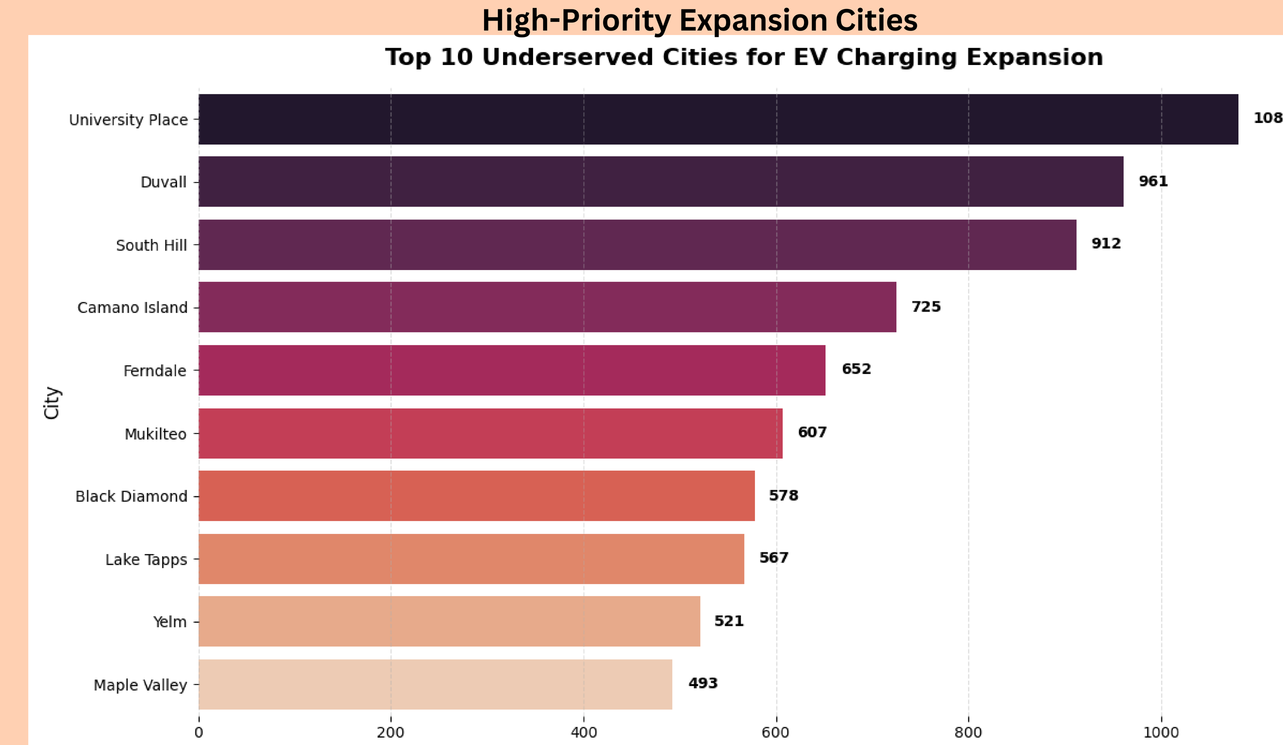
Modeling & Insights

Demand vs Supply Relationship



EV Demand vs Charging Supply

High-Priority Expansion Cities



Top Underserved Cities

Model Used: Linear Regression (OLS) | Dependent: EV Count | Independent: Total Charging Ports

Key Results

- Strong positive relationship between EVs and charging ports
- $R^2 = 0.879$ → model explains ~88% of variation
- Each additional charging port → ~21 more EVs
- Statistically significant (p-value ≈ 0)

Key Insights

- High-demand cities exist with little or no infrastructure
- EV demand is unevenly distributed relative to charging supply
- Some regions show overutilization (congestion risk)
- Others show underutilization (inefficient investment)

EV infrastructure challenge is driven by inefficient distribution, not lack of total supply.

Recommendations

- Expand charging infrastructure into cities with a high EV-per-port ratio
- Primarily focus on high ROI areas where EV demand is strong, but infrastructure is limited
- Potential locations:
  - University Place
  - Duvall
  - South Hill
- Regular check-ins to ensure the company is not overbuilding into saturated markets
- Use infrastructure expansion strategically to drive future EV adoption

Acknowledgments

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